

Pin reference

Proffieboard V3 (incl. 3.9) · ProffieOS

Section **A**: names accepted in SD **config/blades.ini**. Section **B**: full board pin map from **proffieboard_v3_config.h** (buttons, I2C, audio, etc.). GPIO values are MCU indices for V3 only.

Generated 2026-10-06 · SD tokens: 21 · SaberPins: 46

Section A — SD blades.ini tokens

Use these spellings in `data_pin=`, `power_pin*=`, and `simple-blade pin*=` · `blade_config_pin_names.h`

C name	GPIO	Role
blade2Pin	2	PA9 + PB4
blade3Pin	4	PA10 + PB5
blade4Pin	6	PA4
blade5Pin	7	Free1 PB3
blade6Pin	8	Free2 PB10
blade7Pin	9	Free3 PB11 + PC2
blade8Pin	16	also uart PC0
blade9Pin	17	also uart PC1
bladeIdentifyPin	1	blade identify input / FoC
bladePin	0	blade control, either WS2811 or PWM PA7+PA6
bladePowerPin1	20	blade power control PA1
bladePowerPin10	2	PA9
bladePowerPin11	4	PA10
bladePowerPin2	21	blade power control PB6
bladePowerPin3	22	blade power control PC7
bladePowerPin4	23	blade power control PB1
bladePowerPin5	24	blade power control PC6
bladePowerPin6	25	blade power control PB0
bladePowerPin7	7	hook up an external FET to drive more powerful LEDs
bladePowerPin8	8	hook up an external FET to drive more powerful LEDs
bladePowerPin9	9	hook up an external FET to drive more powerful LEDs

Section B — I2C

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C name	GPIO	Role
i2cDataPin	18	I2C bus, Used by motion sensors PB7
i2cClockPin	19	I2C bus, Used by motion sensors PB8

Section B — Buttons

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C name	GPIO	Role
powerButtonPin	11	power button PB14 + PC5
auxPin	13	AUX button PB13 + PA2
aux2Pin	15	AUX2 button PB15

Section B — Memory card now uses SDIO

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C name	GPIO	Role
sdCardSelectPin	255	Memory card now uses SDIO
amplifierPin	32	Amplifier enable pin PH1
boosterPin	31	Booster enable pin PH0
motionSensorInterruptPin	30	motion sensor interrupt PC13

Section B — No fastled support yet

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C name	GPIO	Role
spiLedSelect	-1	No fastled support yet
spiLedDataOut	-1	No fastled support yet
spiLedClock	-1	No fastled support yet

Section B — Neopixel pins

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C name	GPIO	Role
bladePin	0	blade control, either WS2811 or PWM PA7+PA6
bladeIdentifyPin	1	blade identify input / FoC
blade2Pin	2	PA9 + PB4
blade3Pin	4	PA10 + PB5
blade4Pin	6	PA4
blade5Pin	7	Free1 PB3
blade6Pin	8	Free2 PB10
blade7Pin	9	Free3 PB11 + PC2
blade8Pin	16	also uart PC0
blade9Pin	17	also uart PC1

Section B — Blade power control

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C name	GPIO	Role
bladePowerPin1	20	blade power control PA1
bladePowerPin2	21	blade power control PB6
bladePowerPin3	22	blade power control PC7
bladePowerPin4	23	blade power control PB1
bladePowerPin5	24	blade power control PC6
bladePowerPin6	25	blade power control PB0

Section B — hook up an external FET to drive more powerful LEDs

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C name	GPIO	Role
bladePowerPin7	7	hook up an external FET to drive more powerful LEDs
bladePowerPin8	8	hook up an external FET to drive more powerful LEDs
bladePowerPin9	9	hook up an external FET to drive more powerful LEDs

Section B — Status LED

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C name	GPIO	Role
statusLEDPin	26	Status LED

Section B — be possible at 800kHz.

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C name	GPIO	Role
bladePowerPin10	2	PA9
bladePowerPin11	4	PA10

Section B — Analog pins

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C name	GPIO	Role
batteryLevelPin	29	battery level input PC4
chargeDetectPin	27	PA0

Section B — UART

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C name	GPIO	Role
rxPin	16	PC0
txPin	17	PC1

Section B — MiCOM setup

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C name	GPIO	Role
trigger1Pin	11	power button
trigger2Pin	13	aux button
trigger3Pin	15	aux2 button
trigger4Pin	7	free1
trigger5Pin	8	free2
trigger6Pin	9	free3
trigger7Pin	2	data2
trigger8Pin	4	data3

Bonded GPIO pairs (V3)

Do not assign conflicting roles on the same bonded pair: (0,1) (2,3) (4,5) (9,10) (11,12) (13,14)